ShieldSight: PPE Compliance Detection System

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Tier 2 – Object detection

The Problem

Problem

Construction and industrial workers do not always wear the required personal protective equipment (PPE), such as hard hats and safety vests. Manually monitoring every worker can be difficult in large or busy workplaces. Missing PPE can increase the risk of workplace injuries and OSHA violations.

Who Cares?

Safety Managers, Construction Companies, Warehouse Supervisors, and OSHA Compliance Officers.

Why It Matters?

Failure to wear PPE increases the risk of workplace injuries, accidents, OSHA fines, and lost productivity. An automated monitoring system can improve workplace safety while reducing the need for constant manual inspections.

Solution

ShieldSight will automatically detect workers wearing required PPE and highlights potential safety violation using computer vision.

My Solution

Input:

Camera Image → 1 → YOLO11 Object Detection → 2 → Detect Person, Detect Hard Hat, Detect Safety Vest → 3 → Compare PPE Requirements → 4 → Highlight Potential PPE Violations

Final Output: Bounding Boxes, Confidence Scores
PPE Detected, Possible missing PPE highlighted

Technical Approach

Computer Vision Technique

- Object Detection

Model

- YOLO11

Framework

- Ultralytics
- PyTorch
- Google Colab

Why YOLO11?

- YOLO11 is designed for fast, real-time object detection and can detect multiple objects within a single image. The design of YOLO11 makes it a great choice to identifying workers and PPE simultaneously while providing accurate bounding boxes and confidence scores.

Data Plan

Dataset

- Kaggle PPE Detection Dataset

Images

- Approximately **2,000 images**

Labels

- Person
- Helmet/no helmet
- Safety Vest/ no safety vest

Data Preparation

- Training (70%)
- Validation (20%)
- Testing (10%)

Success Metric

Primary Metric Goal – Accuracy atleast 85% or greater

Secondary Metric Goal – Average processing speed < 1 second per image

Additional Goal - Correctly identify workers wearing and missing required PPE in test images

Milestone Plan

Phase	Goal	Milestone	Week
Blueprint	Plan approved	Midterm Submitted	Week 5
First Working Demo	Run pretrained YOLO11 on PPE images	Working Demo	Week 6
Make it yours	Train model using PPE dataset and improve detection	System works on detecting workers and PPE	Week 7-8
Improve & Measure	Test, fix, and measure against success metrics	>=85% accuracy ; process >1 second per image. (Record Metrics)	Week 9
Package & Present	Demo videos, README, final slides	Final Submitted	Week 10

Risks & Resources

Risk 1

- Not enough labeled PPE images.

Plan B

- Use an additional Roboflow or Kaggle dataset.

Risk 2

- Model accuracy is lower than expected.

Plan B

- Use data augmentation, increase training epochs, or fine-tune a larger YOLO11 model.

Compute

- Google Colab

Software

- Python
- Ultralytics
- PyTorch
- GitHub

Estimated Cost - $0